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E102 Spring 2025

Advanced Systems Engineering

Digital Controller Project

Due Date:

Teams with a senior: 5pm Thursday May 8

Teams with juniors: 12pm Monday May 12

Please complete and submit this project workbook detailing the design, simulation, and experimental testing of digital controllers for an overdamped second order LTI system to satisfy the following specifications for the response to a step reference input of 2.5V:

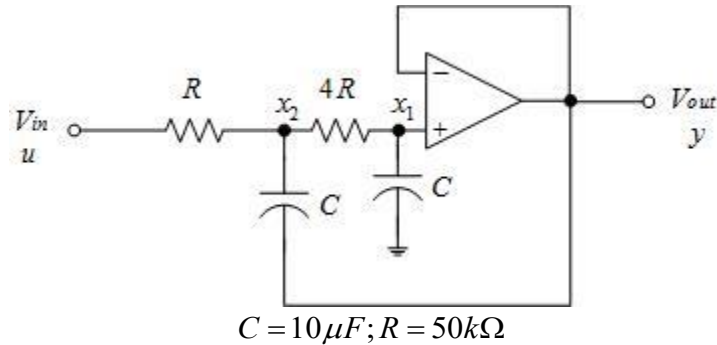
- less than 1% peak overshoot
- zero steady state error
- minimize the control input keeping 1% settling time under 4 s
- sample at 10 Hz
- control input saturates at 5V

Timeline:

1. Team Project posted on Canvas on April 25th.
2. Pick up experimental kit for your team from the engineering stockroom starting April 25th. Complete the “Using the Arduino Uno as a digital controller” tutorial from the class directory.
3. Return kit (disassembled) to the engineering stockroom and submit this Project Workbook on Gradescope for your team by the due date listed above. Make sure you upload only one submission per team on Gradescope (a single team submission on behalf of both partners).

Project Tasks

1. Consider the overdamped second order circuit shown below:



Determine the transfer function $H(s) = \frac{Y(s)}{U(s)}$ ¹

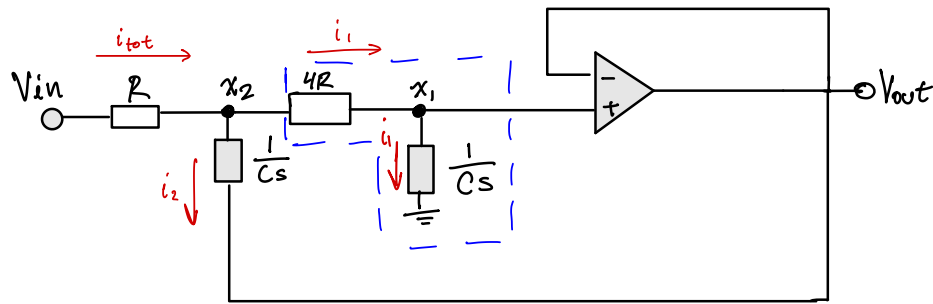
$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{4R^2C^2s^2 + 5RCs + 1}$$

$$H(s) = \frac{1}{4(50,000)^2(10\mu F)^2s^2 + 5(50,000)(10\mu F)s + 1}$$

$$H(s) = \frac{1}{s^2 + 2.5s + 1}$$

¹ All derivations and design calculations should be inserted in the text boxes provided. Handwritten equations are acceptable. **Please indicate final answers with numerical numbers (instead of R and C).**

Determine the transfer function $H(s) = \frac{Y(s)}{U(s)}$



KCL: $i_{tot} = i_1 + i_2$

ideal op-amp laws:
 $V_- = V_+$
 ∞ input impedance

$$V_+ = V_{x_1} = \frac{\frac{1}{Cs}}{(4R + \frac{1}{Cs})} V_{x_2} = \frac{1}{(4RCs + 1)} V_{x_2} \longrightarrow V_{x_2} = \frac{4R + \frac{1}{Cs}}{\frac{1}{Cs}} V_{x_1}$$

$$V_{x_2} = (4RCs + 1) V_{x_1}$$

KCL:

$$\frac{V_{in} - V_{x_2}}{R} = \frac{V_{x_1} - 0}{\frac{1}{Cs}} + \frac{V_{x_2} - V_{out}}{\frac{1}{Cs}}$$

RECALL:

$$V_- = V_+ = V_{x_1} = V_{out}$$

$$\frac{V_{in} - V_{x_2}}{R} = \frac{V_{out}}{\frac{1}{Cs}} + \frac{V_{x_2} - V_{out}}{\frac{1}{Cs}}$$

$$\frac{V_{in} - (4RCs + 1) V_{out}}{R} = Cs V_{out} + \frac{(4RCs + 1) V_{x_1} - V_{out}}{\frac{1}{Cs}}$$

$$= \cancel{Cs V_{out}} + (4RC^2 s^2 + Cs) V_{out} - \cancel{Cs V_{out}}$$

$$= (4RC^2 s^2 + Cs) V_{out}$$

$$V_{in} - (4RCs + 1) V_{out} = (4R^2 C^2 s^2 + RCs) V_{out}$$

$$V_{in} = (4R^2 C^2 s^2 + RCs + (4RCs + 1)) V_{out}$$

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{4R^2 C^2 s^2 + 5RCs + 1}$$

Determine the state space representation $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$
 $y(t) = \mathbf{C}\mathbf{x}(t)$

Use KCL at node x_2 and x_1 to find the voltage across the capacitors:

KCL @ NODE x_2

$$\frac{u - x_2}{R} = C \frac{dx_2}{dt}$$

$$\hookrightarrow \frac{dx_2}{dt} = \frac{u}{RC} - \frac{x_2}{RC}$$

KCL @ NODE x_1

$$\frac{x_2 - x_1}{4R} = C \frac{dx_1}{dt}$$

$$\hookrightarrow \frac{dx_1}{dt} = \frac{x_2 - x_1}{4RC}$$

Now let's write out of state space equation if $y = x_1$:

$$\dot{\mathbf{x}} = \begin{bmatrix} -0.5 & 0.5 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Determine an analytical expression for the step response $y_{step}(t)$

$$y(t) = \frac{5}{2} + \frac{5}{6} e^{-2t} - \frac{10}{3} e^{-\frac{1}{2}t}$$

Determine an analytical expression for the step response $y_{step}(t)$

$$H(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 2.5s + 1}$$

Please complete and submit this project workbook detailing the design, simulation, and experimental testing of digital controllers for an overdamped second order LTI system to satisfy the following specifications for the response to a step reference input of 2.5V:

- less than 1% peak overshoot
- zero steady state error
- minimize the control input keeping 1% settling time under 4 s
- sample at 10 Hz
- control input saturates at 5V

$$u(t) \begin{cases} 2.5V, & t \geq 0 \\ 0, & \text{else} \end{cases}$$

$$\mathcal{L}[1] = \frac{1}{s} \longrightarrow$$

$$\mathcal{L}[2.5] = \frac{2.5}{s} = U(s)$$

$$\rightarrow Y(s) = \frac{1}{s^2 + 2.5s + 1} U(s)$$

$$= \frac{1}{s^2 + 2.5s + 1} \cdot \frac{2.5}{s}$$

$$= \frac{2.5}{s(s^2 + 2.5s + 1)} = \frac{5}{2s(s^2 + 2.5s + 1)}$$

$$= \frac{5}{s(2s^2 + 5s + 2)} = \frac{5}{s(s+2)(2s+1)}$$

$$\left(\text{PFC} \right) \frac{5}{s(s+2)(2s+1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{2s+1}$$

$$5 = A(s+2)(2s+1) + B \cdot s(2s+1) + C(s+2)s$$

for $s=0$:

$$5 = A(s+2)(2s+1) + B \cdot \cancel{s}(2s+1) + C \cancel{(s+2)}s$$

$$5 = A(2)(1) \rightarrow \boxed{A = 2.5}$$

for $s=-2$:

$$5 = A \cancel{(s+2)}(2s+1) + B \cdot s(2s+1) + C \cancel{(s+2)}s$$

$$5 = B \cdot (-2)(2(-2)+1)$$

$$6B = 5 \rightarrow \boxed{B = \frac{5}{6}}$$

for $s=-\frac{1}{2}$:

$$5 = A \cancel{(s+2)}(2s+1) + B \cdot s \cancel{(2s+1)} + C \cancel{(s+2)}s$$

$$5 = C \left(\frac{3}{2} \right) \left(-\frac{1}{2} \right)$$

$$\boxed{C = -\frac{20}{3}}$$

$$Y(s) = \frac{5}{2s} + \frac{5}{6(s+2)} + \frac{-20}{3(2s+1)} \quad \rightarrow \quad -\frac{20}{3} \cdot \frac{1}{2} \cdot \frac{1}{s+\frac{1}{2}}$$

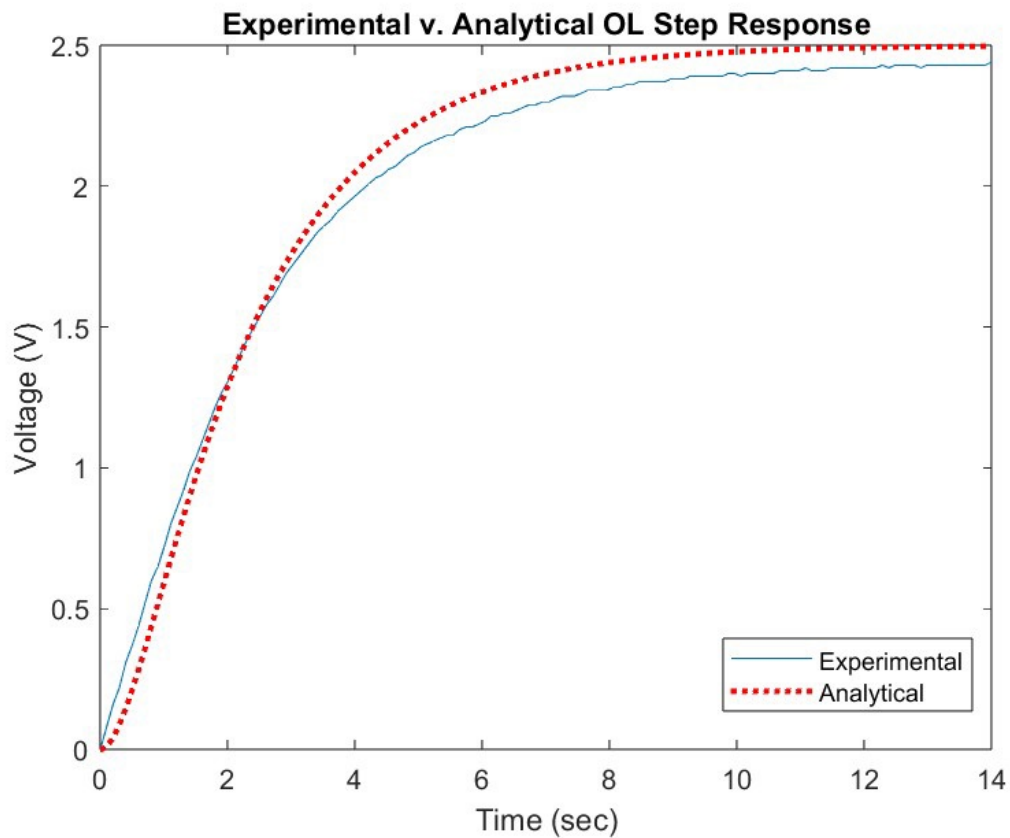
$\downarrow \mathcal{L}^{-1}$

$$\boxed{y(t) = \frac{5}{2} + \frac{5}{6} e^{-2t} - \frac{10}{3} e^{-\frac{1}{2}t}}$$

2. Build the circuit with the LMC6484 op amp (data sheet in class directory) and measure the open loop step response experimentally by coding the Arduino Uno microprocessor to interface with the circuit.

Insert a fully labeled experimental open loop step response plot for a 2.5V step input² with an overlay of the analytical step response from part 1

(insert figure here)



² All plot figures submitted in this workbook must be fully labeled with title, axis labels, and legends for multiple plots. The quality of the figures will be considered a part of the assessment. See the plot examples in the "Using the Arduino Uno as a digital controller" tutorial document

3. Design a unity feedback continuous-time closed loop system for the circuit (plant) with a PI compensator $C(s) = K_p + \frac{K_i}{s}$ for 70° phase margin and crossover frequency $\omega_{co} = 1 \text{ rad/s}$.

Let's go ahead and find the open loop transfer function with the PI compensator as follows:

$$T_{OL}(s) = C(s)H(s) = \frac{K_p + \frac{K_i}{s}}{s^2 + 2.5s + 1} = \frac{K_p(s + \alpha)}{s(s^2 + 2.5s + 1)} \quad * \text{ let } \alpha = \frac{K_i}{K_p}$$

Now we can use our known ϕ_m and ω_{co} values to find K_p and K_i as follows:

$$T_{OL}(j\omega) = \frac{K_p(j\omega + \alpha)}{j\omega((1 - \omega^2) + j\omega 2.5)} \cdot \frac{(1 - \omega^2) - j\omega 2.5}{(1 - \omega^2) - j\omega 2.5} = \frac{K_p(2.5\omega^2 - \alpha\omega^2 + \alpha + j\omega(-\omega^2 - 2.5\alpha + 1))}{j\omega(\omega^4 + 6.25\omega^2 - 2\omega^2 + 1)}$$

$$\angle T_{OL}(j\omega_c) = \tan^{-1} \left(\frac{\omega_c(-\omega_c^2 - 2.5\alpha + 1)}{2.5\omega_c^2 - \alpha\omega_c^2 + \alpha} \right) - 90^\circ = -180^\circ + \phi_m \rightarrow \alpha = 0.36397$$

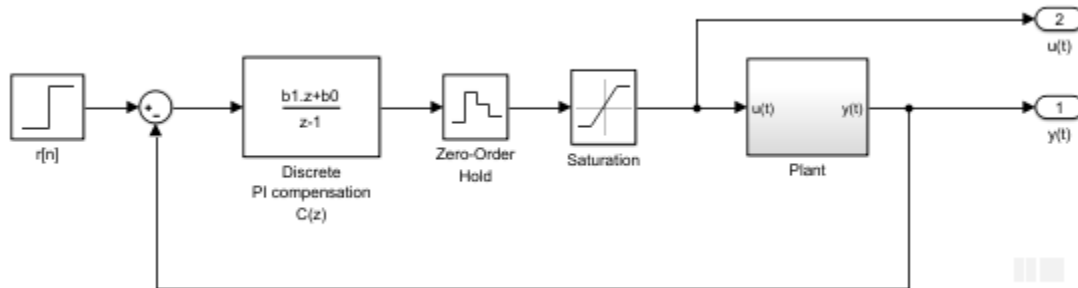
$$|T_{OL}(j\omega_c)| = \frac{K_p \sqrt{(\omega_c^2 + \alpha^2)}}{\omega_c \sqrt{((1 - \omega_c^2)^2 + (\omega_c 2.5)^2)}} \rightarrow K_p \approx 2.3492$$

thus our K_p and K_i values are the following:

$$K_p \approx 2.3492$$

$$K_i = \alpha K_p \approx 0.85504$$

4. Build a SIMULINK model for digital PI control of the circuit with a saturated control $0 \leq u \leq 5V$ and sample time $T = 0.1s$



Simulink => Discrete => Discrete Transfer Fcn. Sample time => T

Simulink => Discrete => Zero-Order Hold. Sample time => T

Emulate (discretize) the PI compensator from part 3 using the backward differencing method.

From part 3, we know that $C(s) = 2.3492 + \frac{0.85504}{s}$. Let's emulate the compensator as follows:

backwards difference method:

$$u'(t) = 0.85504 e(t) + 2.3492 e'(t)$$

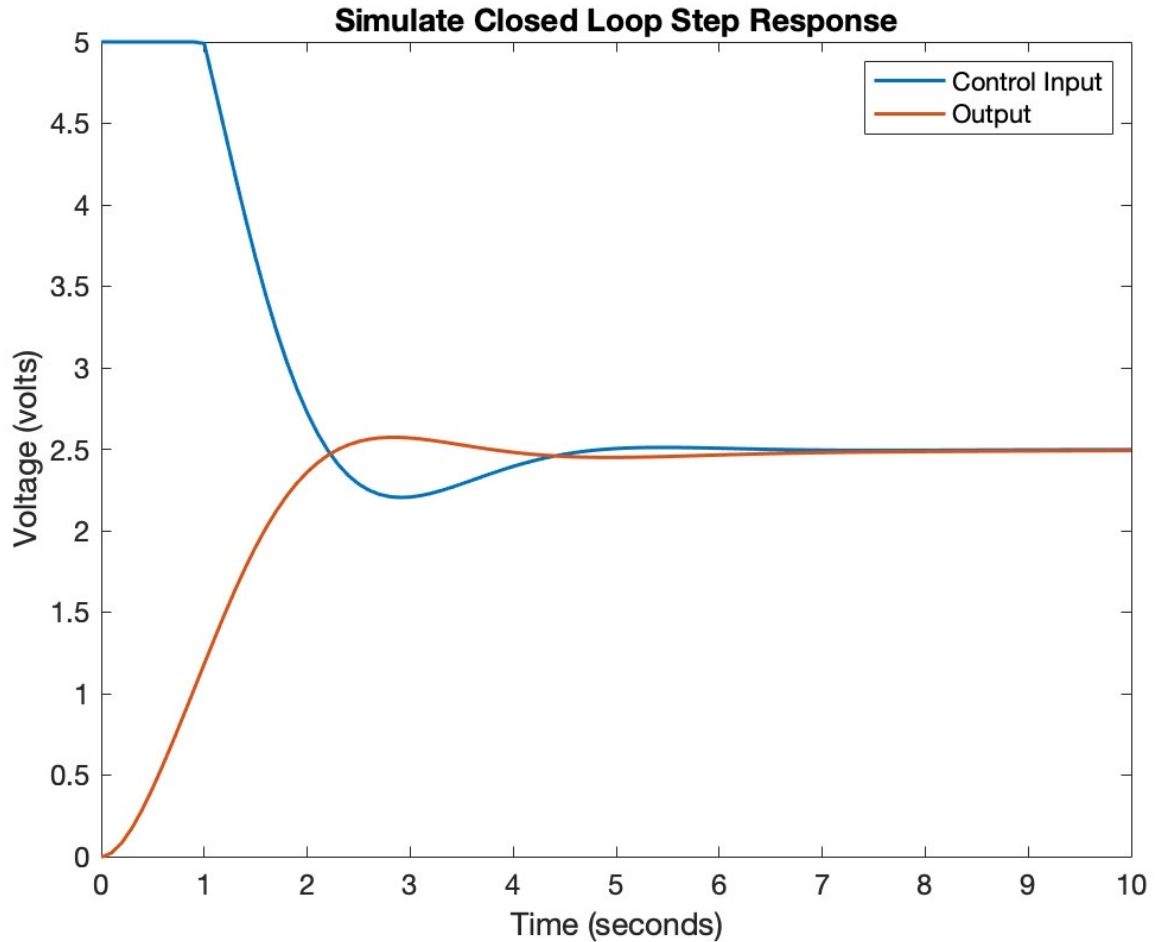
$$\frac{u_n - u_{n-1}}{T} = 0.85504 e_n + 2.3492 \frac{e_n - e_{n-1}}{T}$$

$$u_n = u_{n-1} + e_n \underbrace{[0.85504 T + 2.3492]}_{b_0} + e_{n-1} \underbrace{(-2.3492)}_{b_1}$$

$$C(z) = \frac{2.4347 - 2.3492 z^{-1}}{1 - z^{-1}}$$

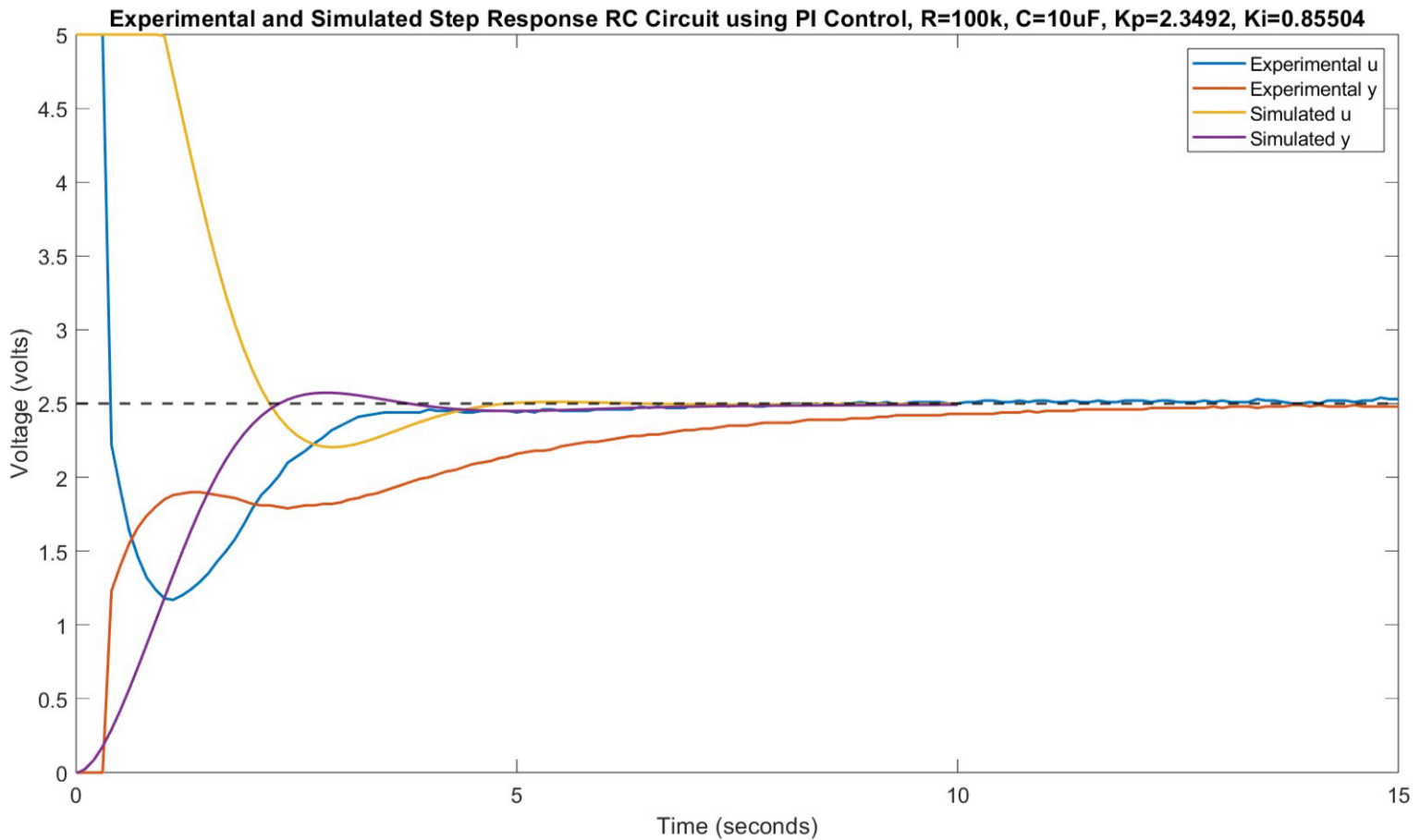
Insert a fully labeled simulated closed loop step response plot of the digital control system for a 2.5V step reference input $r[n]$. Design for a phase margin $\phi_m = 70^\circ$ at crossover frequency $\omega_{co} = 1 \text{ rad/s}$. Plot both the response $y(t)$ and the control input $u(t)$ on the same figure.

(insert figure here)



5. Program the discrete PI closed loop control for the LMC6484 op amp circuit
 Insert a fully labeled experimental closed loop step response plot for a 2.5V step reference input, with an overlay of the simulated step response from part 4.
 Plot both the response $y[n]$ and the control input $u[n]$ on the same figure.

(insert figure here)



CT PI controller: $u(t) = K_p e(t) + K_i \int e(\tau) d\tau$

$u[n] = K_p e[n] + K_i \cdot T_s \sum_{k=0}^n e[k]$

Complete a table of trials of values of design parameters ϕ_m, ω_{co} with the resulting experimental values of 1% settling time $t_{settling}$; peak overshoot M_p ; steady state error e_{ss} and the summation of the absolute deviations of the control input from the final steady state control input $U = \sum_{n=0}^{\infty} |u[n] - u_{ss}|$ where $u_{ss} = \lim_{n \rightarrow \infty} u[n]$.

ϕ_m	ω_{co}	$t_{settling}$	M_p	e_{ss}	$U = \sum_{n=0}^{\infty} u[n] - u_{ss} $
70	1	2.4914	0	0.0094	58.25
80	0.5	2.4737	0	0.0293	15.852
85	2	2.2711	0	0.2311	73.547
90	3	2.444	0	0.0571	86.747
65	2.5	2.3739	0	0.1275	79.6451
75	0.6	2.4934	0	0.0076	28.25

6. Design an observer-based discrete-time state feedback control of the circuit (plant)

- Determine the equivalent discrete-time state space description of the plant for a sample period $T = 0.1s$ Use MATLAB **c2d** command.

$$\begin{aligned} \mathbf{x}[n+1] &= \mathbf{A}_d \mathbf{x}[n] + \mathbf{B}_d u[n] \\ y[n] &= \mathbf{C}_d \mathbf{x}[n] + D_d u[n] \end{aligned} \quad \text{Record } \mathbf{A}_d, \mathbf{B}_d, \mathbf{C}_d, D_d$$

$$\mathbf{A}_d = \begin{bmatrix} 0.9512 & 0.0442 \\ 0 & 0.8187 \end{bmatrix}$$

$$\mathbf{B}_d = \begin{bmatrix} 0.0046 \\ 0.1813 \end{bmatrix}$$

$$\mathbf{C}_d = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D_d = 0$$

%% define continuous ss

```
A = [-0.5 0.5;
      0 -2];
```

```
B = [0;
      2];
```

```
C = [1 0];
```

```
D = [0];
```

```
sysc = ss(A, B, C, D);
```

%% convert continuous ss to discrete

```
sysd = c2d(sysc, 0.1);
```

```
[Ad, Bd, Cd, Dd] = ssdata(sysd);
```

- Design a full state feedback controller to minimize the objective function:

$$J = \frac{1}{2} \sum_{n=0}^{\infty} \left(Q_{11} x_1[n]^2 + Q_{22} x_2[n]^2 + u[n]^2 \right) \quad \text{with } Q_{11} = 100; Q_{22} = 1;$$

Use MATLAB **dlqr** command . Record the full state feedback gain vector $\mathbf{K}$ and the discrete-time closed loop poles

$$\mathbf{K} = \begin{bmatrix} 6.5014 & 1.2697 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 0.7549 + 0.1322j & \\ 0.7549 - 0.1322j & \end{bmatrix}$$

%% calculate feedback gain and closed-loop poles

```
Q = diag([100, 1]);
```

```
R = 1;
```

```
[K,S,P] = dlqr(Ad, Bd, Q, R);
```

- Select the reference gain K_r for zero steady state error to a constant reference input. Record K_r .

$$K_r = 8.7711$$

```
%% calculate reference gain
I = diag([1,1]);
Adcl = Ad-Bd*K;
Kr = 1/(Cd*inv(I-Adcl)*Bd)
```

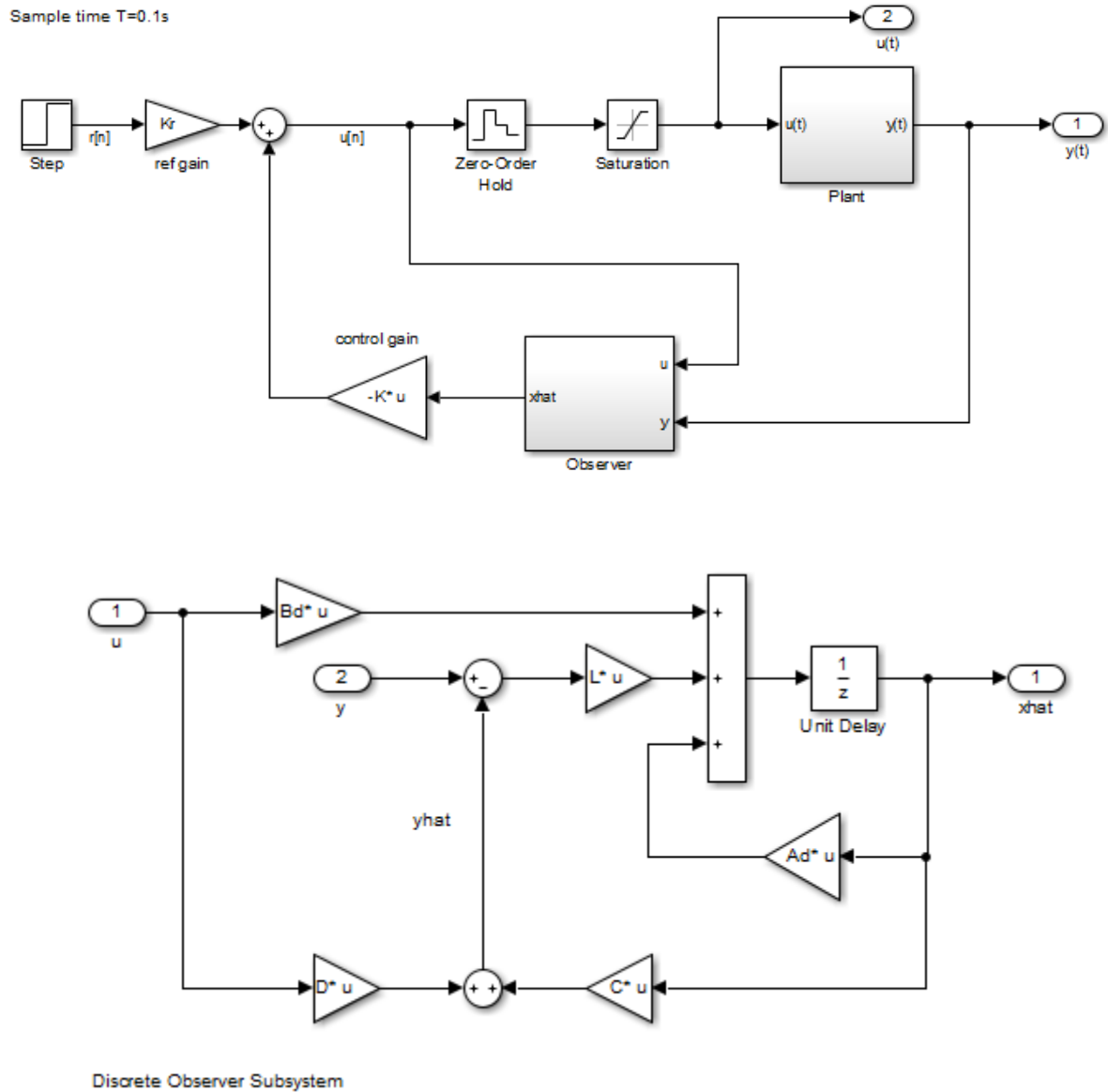
- Design an observer with pole locations at twice the speed of the closed loop poles. Record L and the observer poles.

$$L = \begin{bmatrix} 1.019 \\ 4.5077 \end{bmatrix}$$

```
%% design observer poles
desired_obs_poles = 0.5*P;
L = place(Ad',Cd',desired_obs_poles)'
```

★ we want poles that are closer to the origin, so the observer poles need to be smaller than the controller poles.

7. Build a SIMULINK model for observer-based state feedback control of the circuit with a saturated control $0 \leq u \leq 5V$

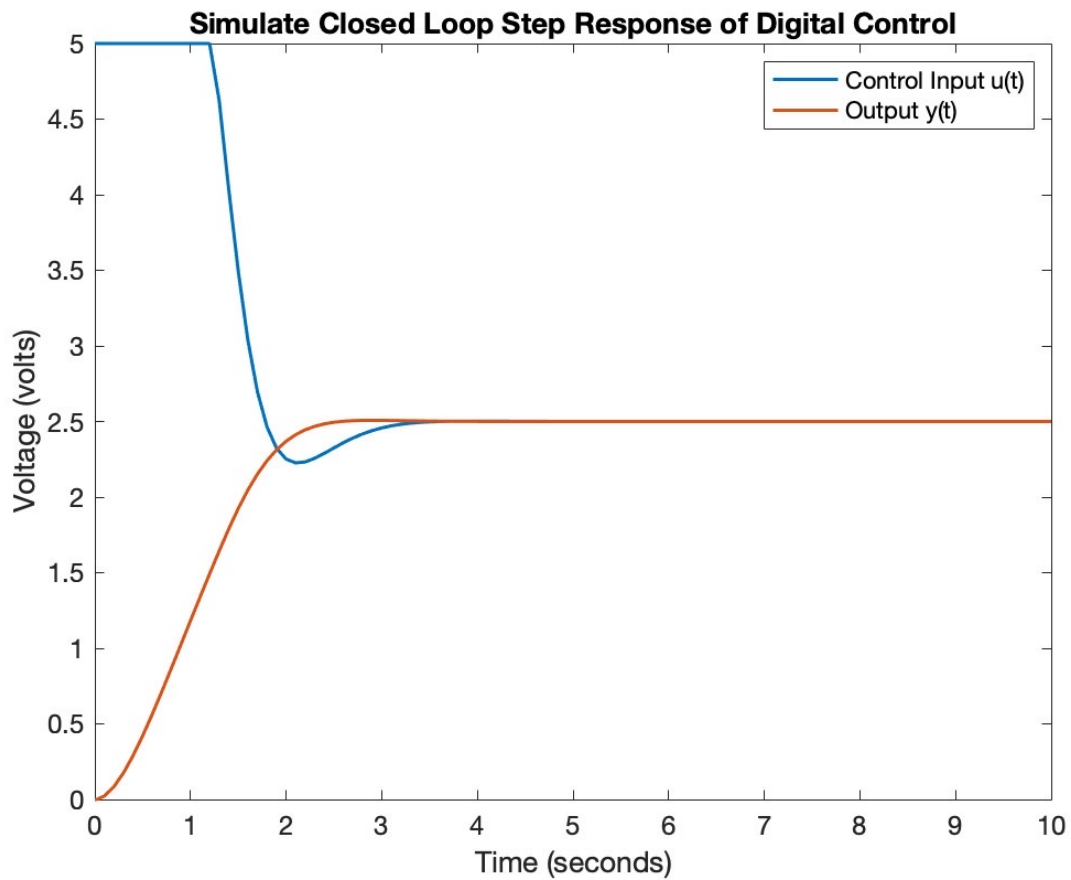


Simulink => *Gain* => *Multiplication* => *Matrix*($K*u$)

Simulink => *Discrete* => *Unit Delay*. *Sample time* => T

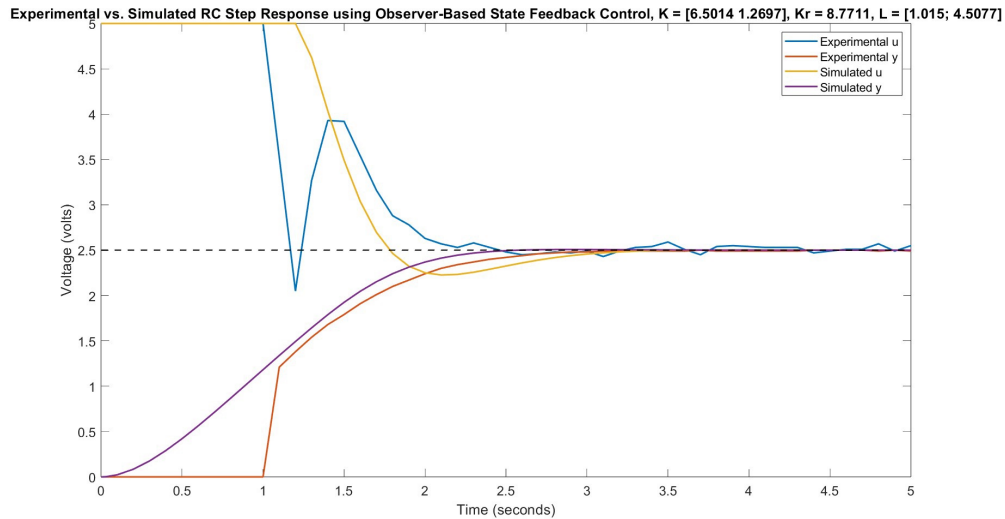
Insert a fully labeled simulated closed loop step response plot of the digital control system for a 2.5V step reference input $r[n]$. Plot both the response $y(t)$ and the control input $u(t)$ on the same figure.

(insert figure here)



8. Program the observer-based state feedback control for the LMC6484 op amp circuit. Insert a fully labeled experimental closed loop step response plot for a 2.5V step reference input, with an overlay of the simulated step response from part 7. Plot both the response $y[n]$ and the control input $u[n]$ on the same figure.

(insert figure here)



$$\hat{x}[n+1] = A_d \hat{x}[n] + B_d u[n] + L(y[n] - \hat{y}[n])$$

$$\hat{y}[n] = C \hat{x}[n] + D u[n]$$

↓ matrix operations are
linear combinations

$$\hat{x}_1[n+1] = a_{11} \hat{x}_1[n] + a_{12} \hat{x}_2[n] + b_1 u[n] + L_1(y[n] - \hat{y}[n])$$

$$\hat{x}_2[n+1] = a_{21} \hat{x}_1[n] + a_{22} \hat{x}_2[n] + b_2 u[n] + L_2(y[n] - \hat{y}[n])$$

$$\hat{y} = C \hat{x}_1[n] + C x_2[n] + D$$

$$A_d = \begin{bmatrix} 0.9512 & 0.0442 \\ 0 & 0.9187 \end{bmatrix}$$

$$K = [6.5014 \quad 1.2697]$$

$$B_d = \begin{bmatrix} 0.0046 \\ 0.1813 \end{bmatrix}$$

$$C_d = [1 \quad 0]$$

$$D_d = 0$$

$$L = \begin{bmatrix} 1.015 \\ 4.5077 \end{bmatrix}$$

$$K_r = 8.7711$$

State variable feedback:

$$u = -Kx + K_r \cdot r$$

$$u = -K\hat{x} + K_r \cdot r$$

$$u[n] = -K\hat{x} + K_r \cdot r[n]$$

Complete a table of trials of values of design control parameters Q_{11}, Q_{22} with the resulting experimental values of 1% settling time t_{settling} ; peak overshoot M_p ; steady state error e_{ss} and the summation of the absolute deviations of the control input from the final steady state control input $U = \sum_{n=0}^{\infty} |u[n] - u_{ss}|$ where $u_{ss} = \lim_{n \rightarrow \infty} u[n]$.

Q_{11}	Q_{22}	t_{settling}	M_p	e_{ss}	$U = \sum_{n=0}^{\infty} u[n] - u_{ss} $
100	1	2.500	0	$-2.7571 \cdot 10^{-11}$	55.0344
1	100	2.4797	0	0.0183	22.4758
100	100	2.4970	0	0.0026	35.7398
50	50	2.4970	0	0.0026	35.5771
25	50	2.4927	0	0.0065	30.6934
25	25	2.4969	0	0.0027	35.3361
25	1	2.5	0	$6.9849 \cdot 10^{-9}$	44.4919
1	1	2.4924	0	0.0067	22.7545

9. Insert a short (1 page) discussion of your rationale for the choice of the control parameters and the effect of your choices on the experimental results

To evaluate our controller design, we tested six configurations by varying the phase margin ϕ_m and crossover frequency ω_{co} , shown in Table 1. We compared each trial based on how well it met the project goals: settling time under 4 seconds, no overshoot, minimal steady-state error, and low control effort.

ϕ_m	ω_{co}	$t_{settling}$	M_p	e_{ss}	$U = \sum_{n=0}^{\infty} u[n] - u_{ss} $
70	1	2.4914	0	0.0094	58.25
80	0.5	2.4737	0	0.0293	15.852
85	2	2.2711	0	0.2311	73.547
90	3	2.444	0	0.0571	86.747
65	2.5	2.3739	0	0.1275	79.6451
75	0.6	2.4934	0	0.0076	28.25

Table 1: Trials of PI design parameters ϕ_m and ω_{co}

Trial 6, with $\phi_m = 75^\circ$ and $\omega_{co} = 0.6$ rad/s, gave the best overall results. It had a settling time of 2.4934 s, well within the requirement, and no overshoot. Most importantly, it had the lowest steady-state error of all trials at 0.0076, showing strong tracking performance. Its control effort was also relatively low at 28.25, making it both accurate and efficient.

While Trial 2 had the lowest control effort (15.852), it came at the cost of higher steady-state error (0.0293), meaning the output didn't quite reach the desired value. Trial 3 settled quickly at 2.2711 s, but its error was unacceptably high (0.2311), making it unreliable. Trial 4 kept good accuracy but required excessive input effort (86.747), which could lead to actuator issues.

Trials 1 and 5 showed solid settling times and low overshoot, but their errors and control effort weren't as well balanced as Trial 6. Overall, Trial 6 hit the sweet spot between performance and efficiency.

These results show that a moderate phase margin and low crossover frequency can lead to stable, accurate, and energy-efficient control. Based on this data, Trial 6 is the recommended setup for implementation—it meets all system requirements and performs well across all key metrics.

10. Insert a short (1 page) discussion of the sources of error between the simulated and experimental step response results

In simulation, components such as resistors and capacitors are assumed to have exact nominal values (e.g. 50 k Ω resistors, 10 μ F capacitors). However, real world components have tolerances that can alter the system's dynamics. For instance, a 5% deviation in the resistor values could shift the circuit's time constants, affecting the settling time or overshoot. Additionally, the LMC6484 op-amp introduces non-ideal behaviors that are not accounted for in the Simulink model. These imperfections manifest into small steady-state errors or unexpected transient behavior in the experimental output.

Additionally, our experimental setup was very susceptible to noise from various sources. The use of jumper cables in the setup that were very easy to wiggle and pull out of the connections likely made a significant difference in the measured responses, since they were able to move around pretty freely during testing. Given that the simulation is also an ideal and simplified representation of an isolated LTI system, there are many factors that it simply can not account for. Apart from electromagnetic interference and ambient noise, there is also the impact due to parasitic capacitance and nonlinearity that are directly linked to the components and possibly even the Arduino Uno itself. The simulation also assumes perfect discretization with a zero-order hold (ZOH) at exactly 10 Hz. As we experienced when gathering experimental data, the Arduino's sampling was less than ideal, which produced artifacts and stuttering that is shown in all of our experimental plots. These effects are absent in the simulation, where everything is computed with basically perfect precision.

All of the factors mentioned above could have also affected the response of the observed-based state feedback controlled system. While the observer-based controller performs well in simulation, the observer's reliance on accurate state estimates makes it sensitive to sensor noise and model inaccuracies. Since the system's actual dynamics differ slightly from what is expected, the observer may introduce noise into the control signal. This may have affected the smoothness of the experimental control input in our plot from part 8.

Despite these discrepancies, the experimental results shown above align well enough with the simulation. While the control input signal seems to noticeably vary from the simulated results, the output signal seems to follow the behavior exhibited by the simulation.